

ACCHUTHA K S

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PROFILE

Highly motivated graduate student pursuing an MCA at Chanakya University with a strong foundation in Machine Learning, Bioinformatics, and Biotechnology. Proven technical expertise through academic research in medical diagnostic modeling and active participation in competitive hackathons. Committed to leveraging data science and AI to solve complex real-world challenges in healthcare and engineering.

EDUCATION

Chanakya University | Bangalore, India

Master of Computer Applications (MCA) | 2025 – Present

- Relevant Coursework: Machine Learning Classifiers, Data Structures and Algorithms (DSA), System Administration, Bioinformatics & Biotechnology.
- Activities: Build for Bengaluru Hackathon.

Chanakya University | Bangalore, India

M.Sc. Bioinformatics and Biotechnology | 2026

ACADEMIC PROJECTS

Comparative Analysis of ML Classifiers for Breast Cancer Diagnosis

Research Project | Chanakya University

- Evaluated and compared five machine learning models (ANN/MLP, Logistic Regression, KNN, Decision Tree, Gaussian Naive Bayes) using the Wisconsin Diagnostic Breast Cancer (WDBC) dataset.
- Implemented a feedforward neural network (ANN) with three hidden layers (500-500-500 neurons) using ReLU activations, achieving a peak accuracy of **99.04%**.
- Integrated **Explainable AI (XAI)** techniques, including Pearson correlation heatmaps and permutation feature importance, identifying 'worst concave points' and 'worst radius' as key clinical predictors.
- Executed a standardized preprocessing pipeline involving z-score normalization (StandardScaler) and a 70/30 train-test partition.

Retro 8-Bit Asteroid Game

Software Development Project

- Designed and built a playable retro arcade game using HTML and JavaScript.
- Developed core game logic for triangular ship movement, collision detection, and asteroid spawning.

TECHNICAL SKILLS

- **Programming:** Python (Advanced), Java (Intermediate), HTML/CSS, JavaScript.
- **Machine Learning & AI:** ANN/MLP, Scikit-Learn, Pandas, NumPy, XAI (SHAP, LIME), Feature Scaling, Data Visualization (Matplotlib, Seaborn).
- **Tools & Systems:** Linux System Administration, Jupyter Notebook, Git/GitHub, Microsoft Office Suite.
- **Domain Expertise:** Bioinformatics, Biotechnology, Medical Diagnostic Modeling, e-commerce data analysis (Real-Time Amazon Data API).

CERTIFICATIONS & TRAINING

- Fundamental Foundations - Machine Learning | Unlox Academy
- 30-Day Linux Programming Roadmap Completion
- Data Structures and Algorithms (DSA) Intensive Training (18 Topics)

STRENGTHS

- Strong analytical reasoning and scientific research capabilities.
- Adaptability to emerging AI, Cloud, and Biotechnology trends.
- Effective collaborator with experience in competitive technical environments (Hackathons).